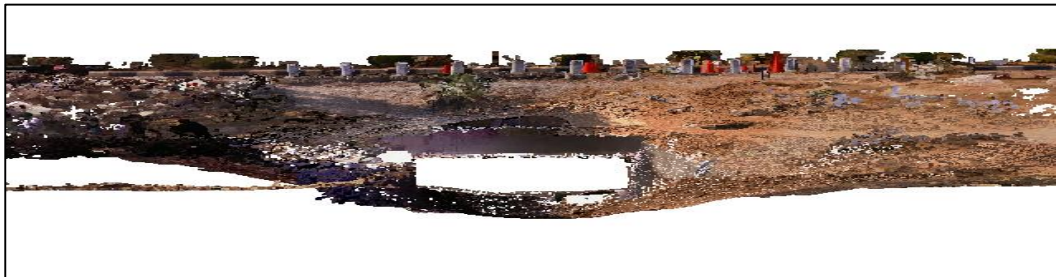


# LIDAR Inspection Report of **Slab Culvert** ch.956+350

**Bridge No:** Slab Culvert

**Bridge Chainage:** 956 + 350 (CKRP)

**Latitude/Longitude:** 25.057348, 74.993851



➤ **Bridge Details**

- **Bridge No:** -
- **Chainage:** 956 + 350
- **Structure type:** Slab Culvert

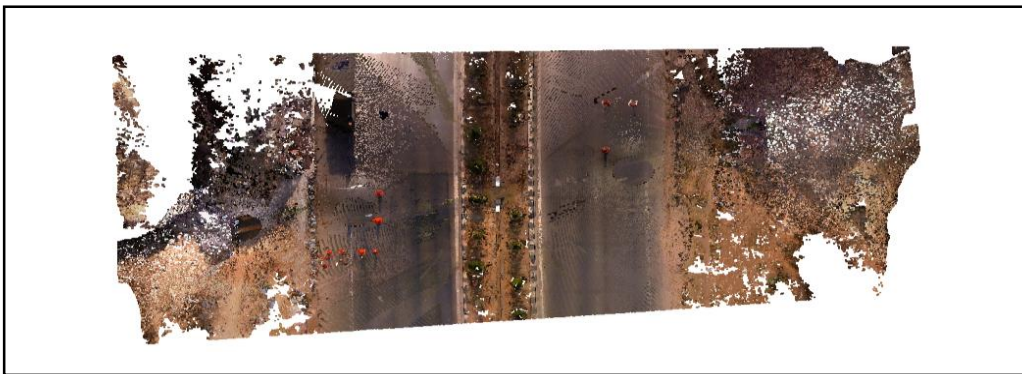
➤ **Project Details**

- **CKRP:** Chittorgarh to kota
- **State:** Rajasthan NH 27
- **Zone:** RJ-06

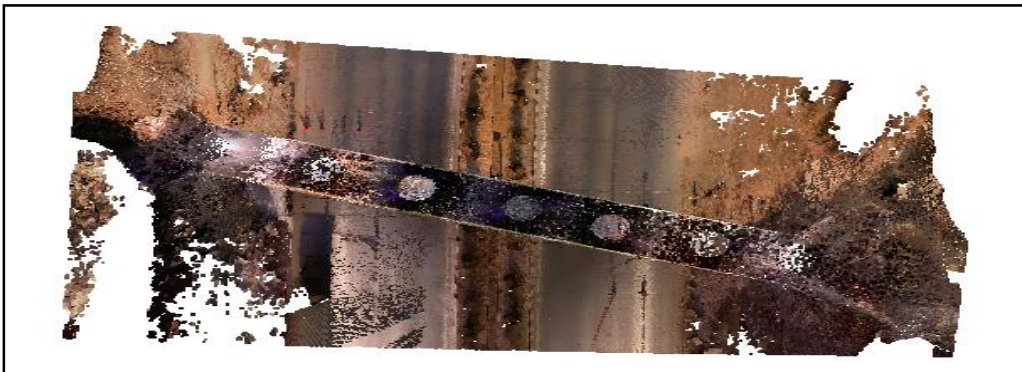
## LiDAR View:



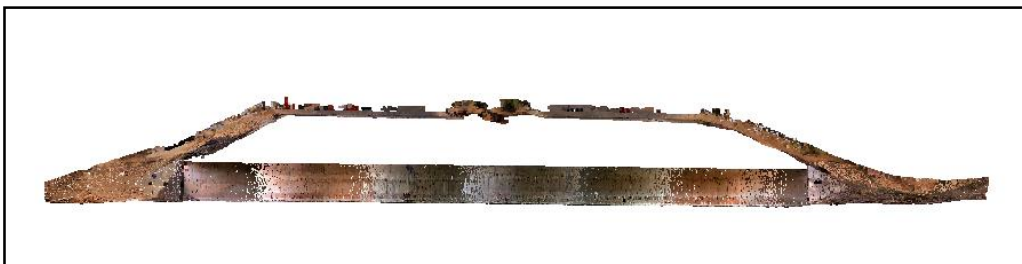
Side View



Top View



Bottom View



Front View

## Structural Measurements

### 1. Height

| Measurement             | Value  |
|-------------------------|--------|
| Abutment bottom to Road | 5.40 m |
| Ground level to slab    | 2.05 m |

### 2. Length

|     | Abutment to<br>Abutment |
|-----|-------------------------|
| LHS | 3.88 m                  |
| RHS | 3.88 m                  |

Deck Width (LHS): 9.60 m

Deck Width (RHS): 9.60 m

## Technical Scope of LiDAR Inspection

| Category                                      | Details                                                                                                                                                    |
|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1. Accuracy Setup and Scan Planning</b>    |                                                                                                                                                            |
| Equipment Deployed                            | <b>Trimble X9</b> – High-performance terrestrial laser scanner with automatic calibration, IP65-rated rugged build, and in-field registration capability   |
| Point Cloud Density                           | Configured for high-density capture with sub-centimeter resolution suitable for fine structural detailing                                                  |
| Measurement Accuracy                          | <b>±1 mm</b> at close range (<60 m); optimal for bridge spans, elevated members, and facia with scan range up to <b>150 meters</b>                         |
| Coverage Strategy                             | Comprehensive planning to prioritize critical structural components including abutments, piers, bearings, and deck; ensured overlap to minimize occlusion  |
| <b>2. Field Setup and Scanning Approach</b>   |                                                                                                                                                            |
| Site Configuration                            | Multiple scan positions arranged for 360° coverage including underside views; referenced globally using <b>GNSS/RTK and Ground Control Points (GCPs)</b>   |
| Execution Environment                         | Scans executed during low-traffic, clear weather conditions to minimize external noise and vibration-related scan distortion                               |
| Safety Measures                               | Operational safety ensured by advance site clearance, barricading, and coordination with local authorities for traffic management                          |
| <b>3. Data Resolution and Quality Control</b> |                                                                                                                                                            |
| Resolution Application                        | Utilized Trimble X9's high-resolution scan settings for capturing large-span structures, vertical components, and minor surface irregularities             |
| Gap Management                                | Conducted supplementary scans from alternate angles; point cloud refinement and merging handled in <b>Trimble RealWorks</b> and <b>Trimble Perspective</b> |
| Data Validation                               | Outlier filtering and noise reduction applied; <b>registration error maintained &lt;5 mm RMSE</b> for consistent dimensional reliability                   |

### Equipment Specifications:

| Feature                      | Trimble X9                                      |
|------------------------------|-------------------------------------------------|
| Scan Speed                   | 1,000,000 pts/sec                               |
| Time to Complete a Full Scan | 4-6 minutes per setup (depending on resolution) |
| Point Cloud Accuracy         | ±3 mm at 20m                                    |
| Maximum Scanning Range       | 150m                                            |
| Field Workflow               | Auto-calibration                                |
| Imaging                      | Standard HDR                                    |
| Durability                   | IP55                                            |

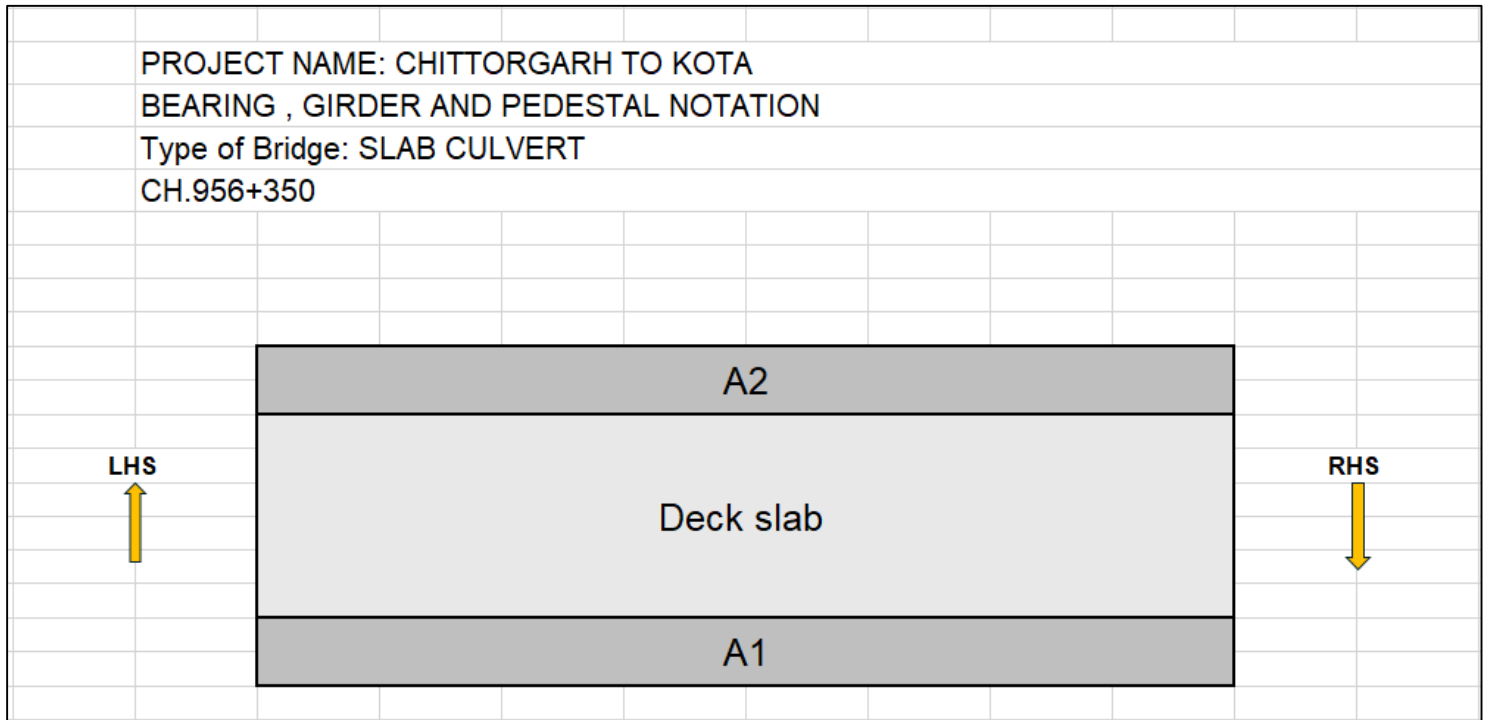
## Table of Content

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## 0. Phase 1 – Scope of Deliverables

| Work Stage                     | Scope / Deliverables                                                     | Location in Report                          |
|--------------------------------|--------------------------------------------------------------------------|---------------------------------------------|
| Accuracy Setup & Scan Planning | Specification of accuracy levels, point cloud density, and scan coverage | Technical Scope of LiDAR Inspection         |
| Technical Setup                | Preparation for LiDAR equipment deployment                               | Technical Scope of LiDAR Inspection         |
| Objective Framing              | Component prioritization for scanning                                    | Structural Measurements + Observation Sheet |
| Field Data Collection          | LiDAR data acquisition                                                   | Structure Layout + Photos                   |
| Resolution Analysis            | Data density and resolution analysis                                     | Technical Scope of LiDAR Inspection         |
| Gap Filling                    | Quality enhancement of point cloud                                       | Embedded in Visual/Photo Section            |
| Point Cloud Analysis           | Structural details and interpretation                                    | Observation Sheet + Distress Photos         |

## 1. Structure Layout



## 2. Layout Legends

| S. No | Description               | Notation |
|-------|---------------------------|----------|
| 1     | Abutment                  | AB       |
| 2     | Foundation of Abutment    | AF       |
| 3     | Abutment Cap              | AC       |
| 4     | Bearing Pedestal          | BP       |
| 5     | Bearings                  | BR       |
| 6     | Foundation of Pier        | PF       |
| 7     | Pier                      | PR       |
| 8     | Pier Cap                  | PC       |
| 9     | Span                      | SP       |
| 10    | Slab span                 | SL       |
| 11    | Box Girder span           | BG       |
| 12    | I Girder                  | IG       |
| 13    | Rectangular Girder        | RG       |
| 14    | Square Girder             | SG       |
| 15    | T Beam Girders            | TB       |
| 16    | Concrete Deck             | CC       |
| 17    | Bituminous Deck           | BD       |
| 18    | Slab b/n girders          | BS       |
| 19    | Cross Girders             | CG       |
| 20    | Railing LHS               | RL       |
| 21    | Railing RHS               | RR       |
| 22    | Footpath LHS              | FL       |
| 23    | Footpath RHS              | RR       |
| 24    | Expansion Joints          | EJ       |
| 25    | Pourable/Crack Joint Seal | PJ       |
| 26    | Compression Joint Seal    | CJ       |
| 27    | Assembly Joint Seal       | AJ       |
| 28    | Strip Seal Joint          | SJ       |
| 29    | Drainage Spouts           | DS       |
| 30    | Wing Wall                 | WW       |
| 31    | Reinforced earth wall     | RE       |
| 32    | Approach Slab             | AS       |

### 3. Distress Nomenclature

| Legend | Description                               | Legend    | Description         |
|--------|-------------------------------------------|-----------|---------------------|
| BG     | Bulging                                   | SC        | Steel Corroded      |
| CRA    | Map Crack/ Crazing                        | SE        | Steel Exposed       |
| RC     | Crack along main reinforcement            | SPE       | Seepage Evident     |
| RP     | Crack perpendicular to Main Reinforcement | SPM       | Seepage Marks       |
|        |                                           | SL        | Scaling             |
| MC     | Minor Crack                               | SG/(G)(S) | Sag in Girder, Slab |
| WC     | Wide Crack                                | EFF       | Efflorescence       |
| SPL    | Spalling                                  | STL       | Settlement          |
| DL     | Delaminated Section                       | H         | Hollow Section      |
| HC     | Honey Combing                             | VEG       | Vegetation Growth   |
| TL     | Tilt                                      | PD        | Pedestal Damage     |

## 4. Observations

**LHS:**

| Element Name  | Observation | Location of Distress | Photo No. | Units | Distress Details |      |       | Area of Repair (Sq.m/R MT/m3) |
|---------------|-------------|----------------------|-----------|-------|------------------|------|-------|-------------------------------|
|               |             |                      |           |       | L(M)             | B(M) | D(M)  |                               |
| Slab          | Damage      | -                    | 1         | M     | 0.99             | 0.15 |       | 0.148                         |
| Abutment Wall | Damage      |                      | 3         | M     | 0.37             | 0.28 | 0.024 | 0.024                         |

**RHS:**

| Element Name  | Observation | Location of Distress | Photo No. | Units | Distress Details |      |      | Area of Repair (Sq.m/R MT/m3) |
|---------------|-------------|----------------------|-----------|-------|------------------|------|------|-------------------------------|
|               |             |                      |           |       | L(M)             | B(M) | D(M) |                               |
| Abutment Wall | Damage      |                      | 2         | M     | 1.41             | 0.18 |      | 0.253                         |

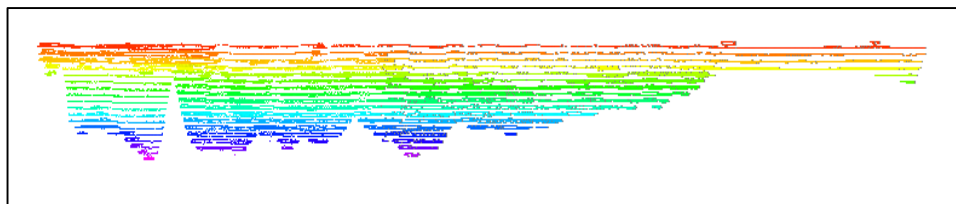
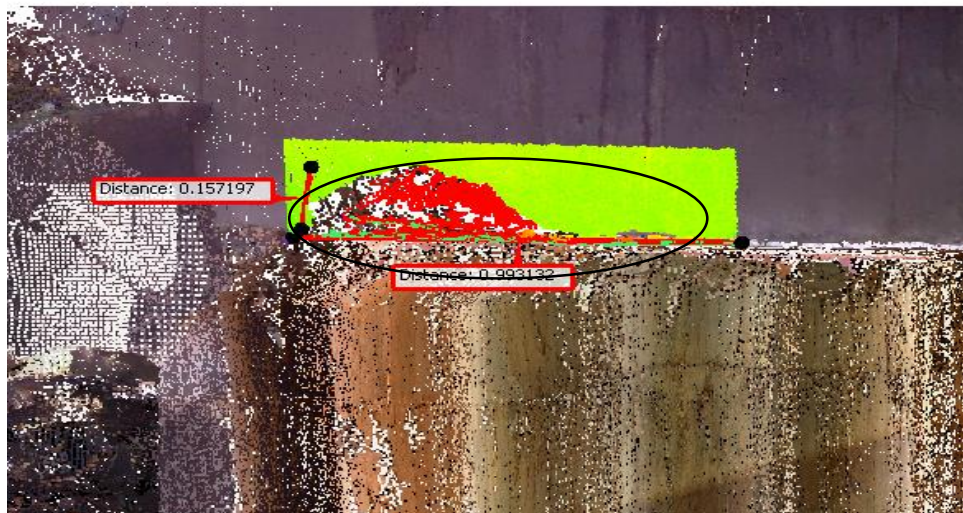
### Observation sheet Non- Structural Elements

| Sl. No. | Description                           | Distress Notation | Observation | Location |       | Distress Details |       |       | Area of Repair |
|---------|---------------------------------------|-------------------|-------------|----------|-------|------------------|-------|-------|----------------|
|         |                                       |                   |             | X (m)    | Y (m) | L (m)            | W (m) | D (m) | Sqm            |
| 1       | Return wall                           |                   | OK          |          |       |                  |       |       |                |
| 2       | Riding return wall                    |                   | NA          |          |       |                  |       |       |                |
| 3       | Wing wall                             |                   | Damage      |          |       |                  |       |       |                |
| 4       | Butterfly walls at top of median wall |                   | NA          |          |       |                  |       |       |                |
| 5       | Dirt wall at top of abutment          |                   | Ok          |          |       |                  |       |       |                |
| 6       | Gabion walls                          |                   | NA          |          |       |                  |       |       |                |
| 7       | Hazard Marker sign board              |                   | OK          |          |       |                  |       |       |                |
| 8       | Hand railing                          |                   | NA          |          |       |                  |       |       |                |
| 9       | Crash barrier                         |                   | NA          |          |       |                  |       |       |                |
| 10      | Median wall                           |                   | NA          |          |       |                  |       |       |                |
| 11      | Pipe Hand Railing (on crash Barrier)  |                   | OK          |          |       |                  |       |       |                |
| 12      | Cleaning of expansion joints          |                   | NA          |          |       |                  |       |       |                |
| 13      | Wheel guard (Mounted & Unmounted)     |                   | NA          |          |       |                  |       |       |                |
| 14      | Footpath                              |                   | OK          |          |       |                  |       |       |                |
| 15      | Drainage Spouts                       |                   | NA          |          |       |                  |       |       |                |
| 16      | Drainage Spout Grating                |                   | NA          |          |       |                  |       |       |                |
| 17      | Drainage Spout Extension Pipe         |                   | NA          |          |       |                  |       |       |                |
| 18      | Weep Holes                            |                   | NA          |          |       |                  |       |       |                |
| 19      | Fascia Panel                          |                   | NA          |          |       |                  |       |       |                |

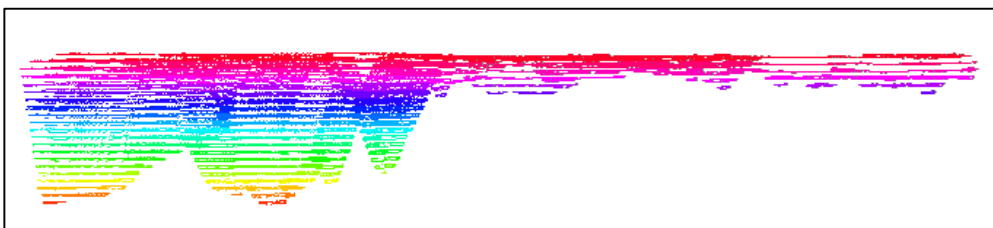
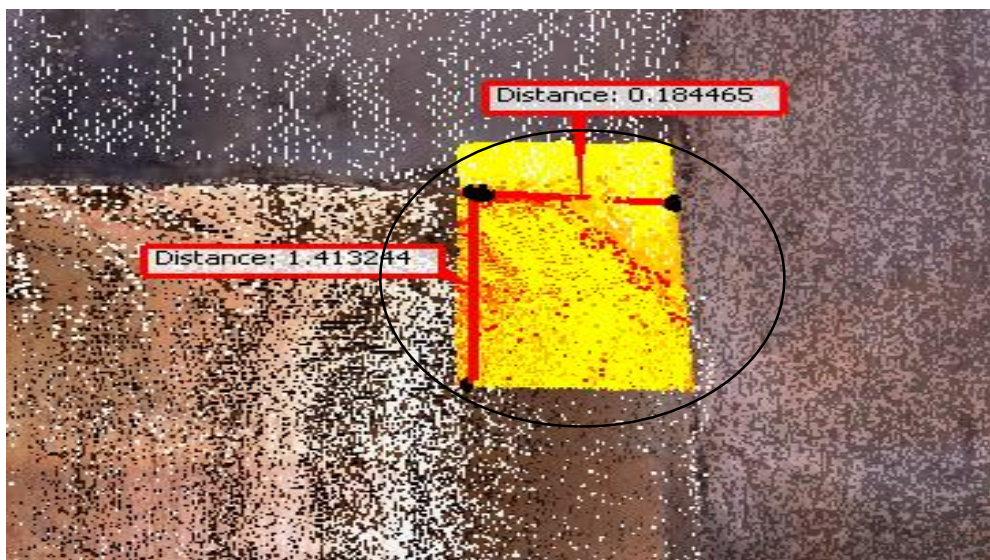
| Sl. No. | Description                      | Distress Notation | Observation | Location |       | Distress Details |       |       | Area of Repair |
|---------|----------------------------------|-------------------|-------------|----------|-------|------------------|-------|-------|----------------|
|         |                                  |                   |             | X (m)    | Y (m) | L (m)            | W (m) | D (m) | Sqm            |
| 20      | Friction Slab                    |                   | Ok          |          |       |                  |       |       |                |
| 21      | Approach slab including PCC      |                   | OK          |          |       |                  |       |       |                |
| 22      | Backfilling below Approach slab  |                   | OK          |          |       |                  |       |       |                |
| 23      | Stone Pitching in quarter coning |                   | NA          |          |       |                  |       |       |                |
| 24      | Filling in quarter coning        |                   | NA          |          |       |                  |       |       |                |
| 25      | Rigid apron                      |                   | NA          |          |       |                  |       |       |                |
| 26      | Curtain wall                     |                   | NA          |          |       |                  |       |       |                |
| 27      | Flexible apron                   |                   | NA          |          |       |                  |       |       |                |
| 28      | Vent clearance                   |                   | NA          |          |       |                  |       |       |                |
| 29      | Toe wall of Quarter coning       |                   | NA          |          |       |                  |       |       |                |
| 30      | RCC drains in tunnel             |                   | NA          |          |       |                  |       |       |                |
| 31      | Cover slab of RCC drains         |                   | NA          |          |       |                  |       |       |                |
| 32      | RCC portal of tunnel             |                   | NA          |          |       |                  |       |       |                |
| 33      | Counter fort wall                |                   | NA          |          |       |                  |       |       |                |
| 34      | Booths structure                 |                   | NA          |          |       |                  |       |       |                |
| 35      | RCC Booth guard                  |                   | NA          |          |       |                  |       |       |                |
| 36      | C.C Panels including PCC         |                   | NA          |          |       |                  |       |       |                |
| 37      | Pipe drain                       |                   | NA          |          |       |                  |       |       |                |
| 38      | Pipe drain cover                 |                   | NA          |          |       |                  |       |       |                |
| 39      | Drain openings                   |                   | NA          |          |       |                  |       |       |                |

## **CH: 956+350 – MNB**

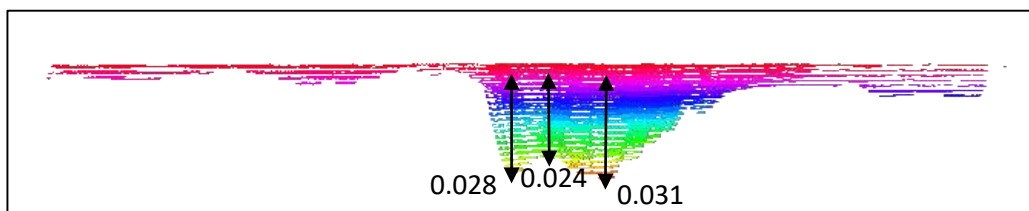
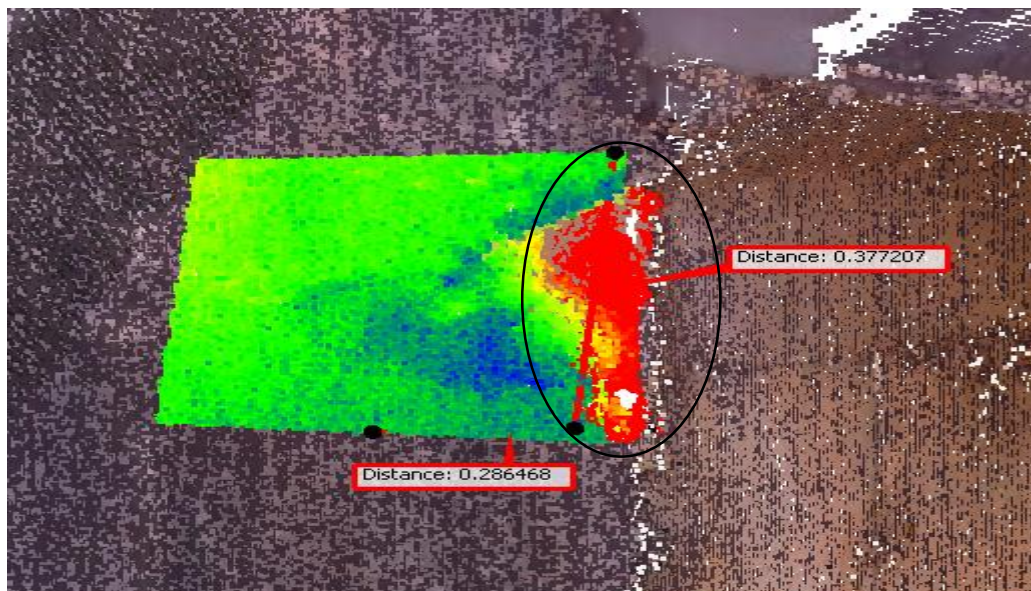
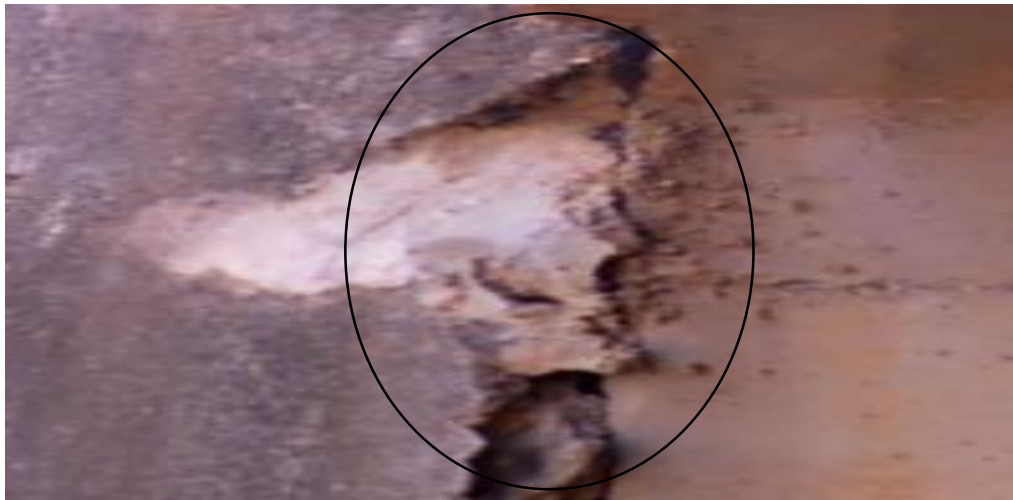
DISTRESS PHOTOS



1. Slab damage observed on LHS, measuring 0.99 m in length and 0.15 m in width, covering an area of 0.148 m<sup>2</sup>.



2. Damage observed on RHS abutment wall, measuring 1.41 m in length and 0.18 m in width, covering an area of 0.253 m<sup>2</sup>.



**3. Damage observed on LHS abutment wall, measuring 0.37 m in length and 0.28 m in width. Depth ranges from a minimum of 0.024 m to a maximum of 0.031 m (average depth = 0.24 m), resulting in a calculated volume of 0.024 m<sup>3</sup>.**